Spatio-Temporal Analysis of Urban Sprawl using Remote Sensing Techniques in Barpeta District

SARANGAPANI BARUAH

3. Methodology (Points – Simple & Crisp)

- The study uses satellite images from **Landsat-8 OLI/TIRS (12 February 2022)** and **Landsat-7 ETM+ (28 January 2000)**.
- Images were downloaded from the **USGS Earth Explorer** platform.
- Landsat data has **30 m spatial resolution**, suitable for analyzing land-cover changes.
- Image processing and classification were performed using **ArcGIS 10.7**.
- Different spectral bands were **stacked to create False Color Composite (FCC)** images.
- **Land Use/Land Cover (LULC)** and **Normalized Difference Built-up Index (NDBI)** were generated.
- These parameters were used to **analyze urban sprawl in Barpeta District**.

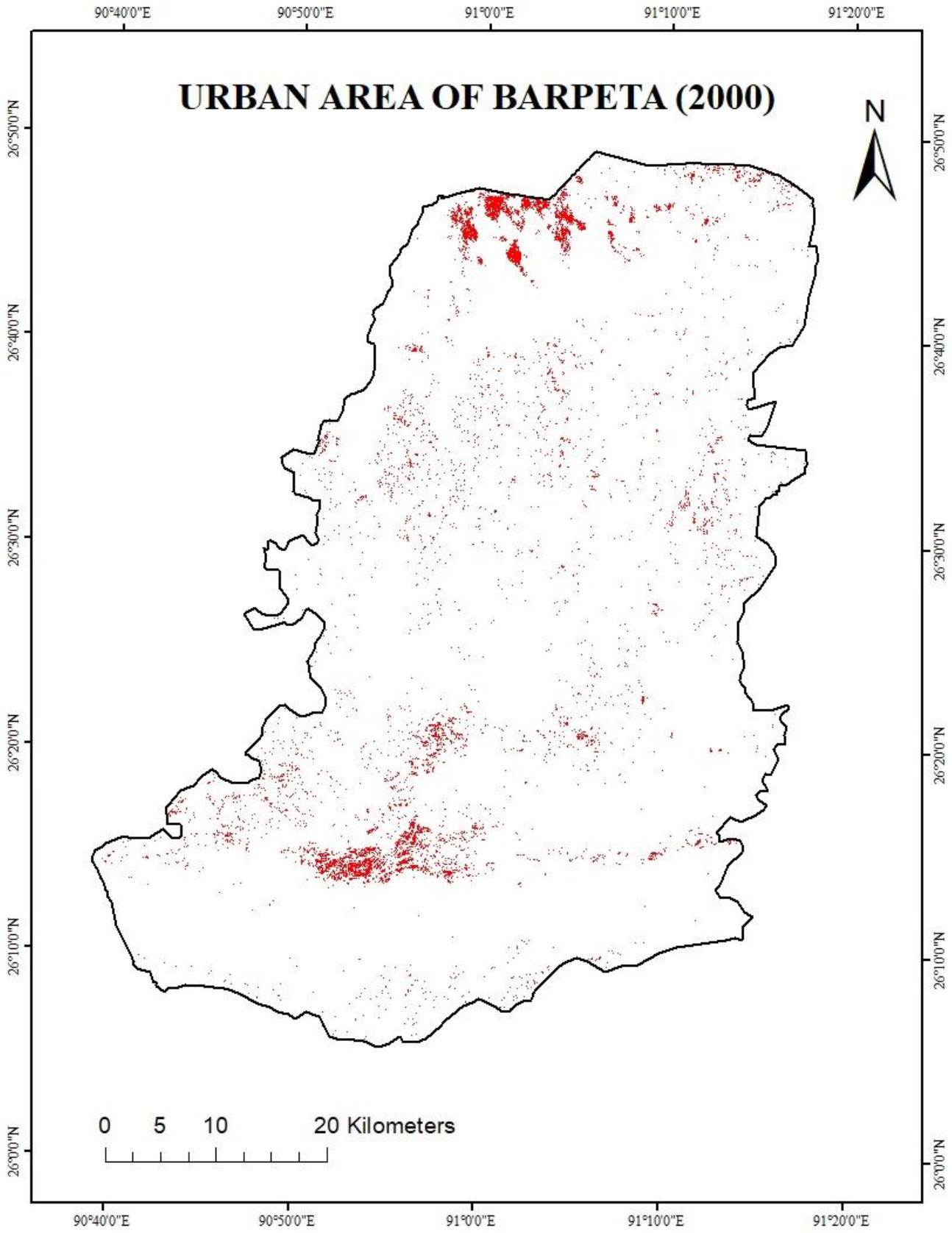

Figure 4.4: Built-up area of Barpeta during 2000

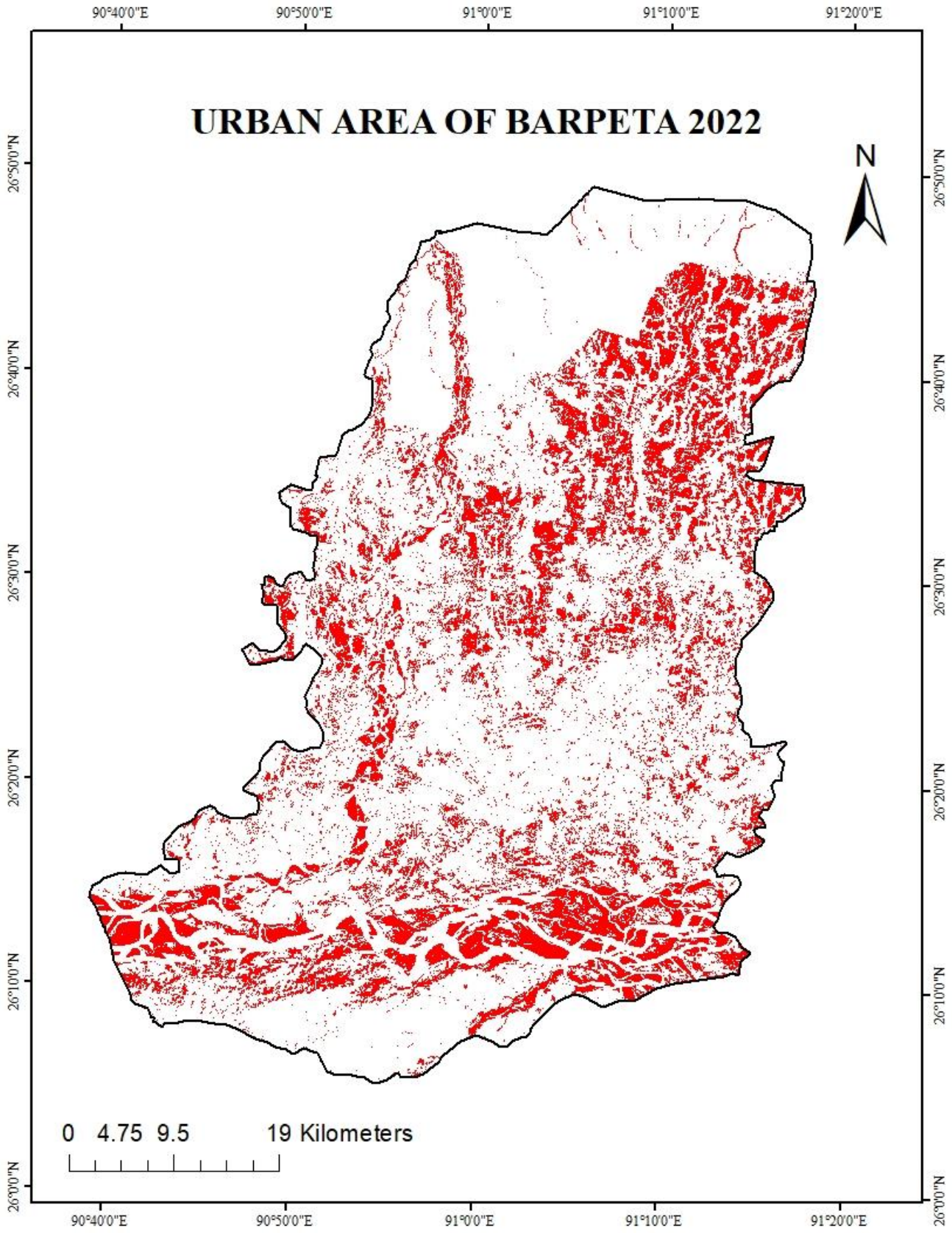

Figure 4.5: Built-up area of Barpeta during 2022